The Unit Circle and Exact Values

Section 1: Coordinates on the Unit Circle

1 On the unit circle, what do the coordinates (x, y) of a point at angle θ represent?

a. $x = \cos\theta$ and $y = \sin\theta$

b. $x = \sin\theta$ and $y = \cos\theta$

c. $x = \tan\theta$ and $y = \cot\theta$

d. $x = r\cos\theta$ where $r > 1$

2 Find the exact coordinates on the unit circle corresponding to $\theta = \frac{5\pi}{6}$.

a. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

3 Find the exact coordinates on the unit circle for $\theta = \frac{4\pi}{3}$.

a. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

b. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

c. $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

Section 2: Reference Angles and Signs by Quadrant

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4 Find the reference angle θ' for $\theta = 225^\circ$.

a. 45°

b. 35°

c. 135°

d. 55°

5 In which quadrants is $\tan\theta$ positive?

a. Quadrants I and III

b. Quadrants I and II

c. Quadrants I and IV

d. Quadrants II and IV

6 If $\cos\theta < 0$ and $\sin\theta > 0$, in which quadrant does the terminal ray of θ lie?

a. Quadrant II

b. Quadrant I

c. Quadrant III

d. Quadrant IV

Section 3: Exact Values of the Six Trig Functions

7 Find the exact value of $\tan\left(\frac{2\pi}{3}\right)$.

a. $-\sqrt{3}$

b. $\sqrt{3}$

c. $-\frac{\sqrt{3}}{3}$

d. -1

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8 Find the exact value of $\csc\left(\frac{7\pi}{4}\right)$.

a. $-\sqrt{2}$

b. $\sqrt{2}$

c. $-\frac{\sqrt{2}}{2}$

d. -2

9 Evaluate $\cos(3\pi) - \sin\left(\frac{3\pi}{2}\right)$.

a. 0

b. -2

c. 2

d. 1

Section 4: Quadrantal Angles and Negative Angles

10 Which of the following values is undefined?

a. $\tan(90^\circ)$

b. $\sin(90^\circ)$

c. $\cos(90^\circ)$

d. $\cot(45^\circ)$

11 Evaluate $\cos(-\pi/3)$.

a. $1/2$

b. $-1/2$

c. $\sqrt{3}/2$

d. $-\sqrt{3}/2$

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12 Evaluate $\sin(-\pi/4)$.

a. $-\frac{\sqrt{2}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. -1

d. 0

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